

Ansatz Comparison Across Dissociation

Where a hardware-efficient circuit stops working, and how to find out why

Phases 0–5 produce a complete and correct project that several thousand people have also produced. Phase 6 produces a result.

The experiment: run the identical scan with a chemistry-motivated ansatz and a generic hardware-efficient one, find where they diverge, and design a diagnostic that distinguishes an expressibility ceiling from an optimiser trap — two failure modes that are indistinguishable from the outside.

The answer is a three-part finding, and it revises a conclusion from Phase 4.

Estimated time: **open** — a focused weekend for the core experiment.

In this document

Two philosophies of ansatz design, and what each guarantees

Expressibility, barren plateaus, and particle-number conservation

A controlled experiment: four conditions across 43 geometries

The diagnostic that separates a structural ceiling from bad luck

Three findings, including one that overturns Phase 4's conclusion about warm-starting

How to write up a result you can defend, and where to go next

1. Where this phase sits

Phases 0–5 are a complete, verified project. They are also a project that several thousand other people have completed, because the H_2 dissociation curve is the standard worked example of the field. Phase 6 is what makes the work **yours**: a genuine experiment, with a result that is not in the tutorials, built entirely from machinery you already have.

The question this phase answers

Phase 3 asserted that ansatz choice is the physics content of VQE, and that a circuit which cannot express the true ground state will converge confidently onto the wrong answer with no warning.

That claim was made and then left untested. **Phase 6 tests it** — by running the same scan, on the same Hamiltonians, with the same optimiser, changing only the circuit.

The answer turns out to be more interesting than "chemistry-motivated ansätze are better", which is the result you might have expected.

If you have built a VQE before

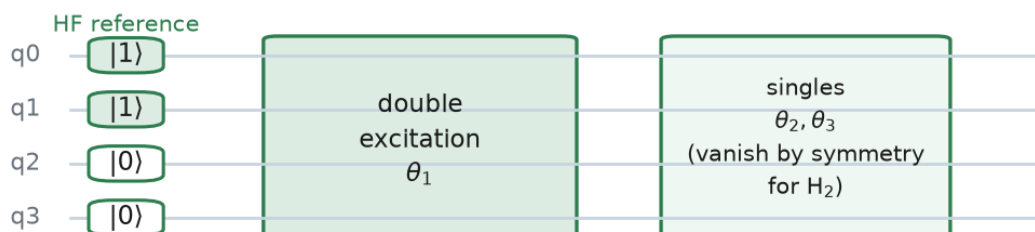
Most introductory VQE demonstrations use a *hardware-efficient* ansatz — Qiskit's `TwoLocal`, or an equivalent stack of rotation and entangling layers — because it is short, generic, and runs on real devices. If that describes something you have written, this phase tells you exactly when the circuit you used stops working, and turns that earlier project into the motivation for this one.

2. Theory

2.1 Two philosophies of ansatz design

	UCCSD	Hardware-efficient
Derived from	Electronic structure theory	What is cheap to run on hardware
Structure	Excitation operators from the orbital space	Alternating rotation and entangling layers
Parameters (H_2 , 4 qubits)	3	24 at 2 repetitions
Conserves particle number	Yes, by construction	No
Circuit depth	Deeper — non-local excitation operators	Shallow, hardware-friendly
Guarantee	Exact for H_2 in a minimal basis	None

UCCSD — chemistry-motivated, 3 parameters



Hardware-efficient (TwoLocal-style) — 24 parameters, no chemical structure

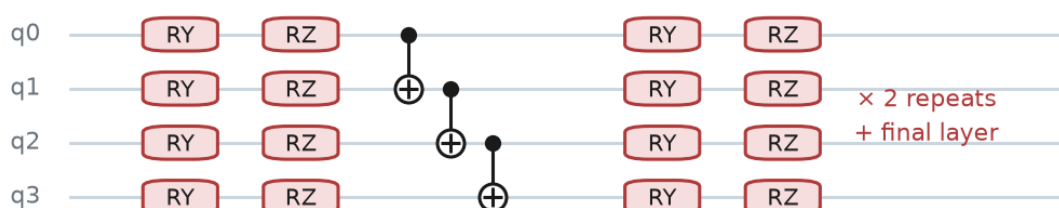


Figure 6.1 — The two circuits. Note that the hardware-efficient one has eight times as many parameters and no chemical structure whatsoever.

2.2 Expressibility

Expressibility is the question of whether the true ground state lies anywhere in the family of states your circuit can produce. If it does not, no optimiser — however good — can find it. The variational principle still holds, so you get a valid upper bound; it is just not a tight one.

A hardware-efficient circuit at fixed depth spans some manifold of states determined by its gate structure. Adding repetitions enlarges that manifold. The question this phase answers empirically is whether two repetitions are enough for H_2 across the whole dissociation curve.

2.3 Barren plateaus and the optimisation landscape

Barren plateau

As qubit count and circuit depth grow, the energy landscape of a randomly initialised hardware-efficient circuit flattens exponentially. Gradients vanish, and the optimiser has nothing to follow. Chemistry-motivated ansätze with good starting points largely avoid this.

Local minimum

A point where the gradient vanishes but the energy is not the lowest achievable. Distinct from an expressibility ceiling: here the state is reachable, but the optimiser did not reach it.

These two failure modes look identical from the outside

Both give you a converged optimiser and an energy that is too high. Distinguishing them requires an experiment, not inspection — and designing that experiment is the most valuable thing in this phase.

The tempting diagnostic is to run many random restarts and see whether they scatter or agree. Section 4.2 shows that this is not enough on its own, and can point you at the wrong conclusion entirely.

The decisive one is a control: reach the same geometry by a path the optimiser can actually follow, rather than by a random jump. If a guided route succeeds where every random start failed, the circuit was expressive enough all along.

2.4 Particle-number conservation

UCCSD is built from excitation operators, every one of which moves electrons without creating or destroying them. It therefore cannot leave the two-electron sector. A hardware-efficient circuit has no such constraint: `RY` and `CNOT` know nothing about electrons, so the state can drift into sectors with the wrong particle number.

For H_2 this is harmless, and Phase 2 established why — exact diagonalisation over all sectors gives the same -1.137306 Ha as the two-electron sector alone, so the chemically meaningful state is the global ground state. For larger molecules this is not guaranteed, and an unconstrained ansatz can converge onto a state with the wrong number of electrons that happens to be lower in energy.

3. Build guide

Step 6.1 — The hardware-efficient ansatz

Everything else already exists. This is the only new component.

Listing 6.1 — the hardware-efficient circuit

```
def build_he(n_qubits, reps=2, electrons=ELECTRONS):
    """Hardware-efficient ansatz: RY/RZ rotations plus a CNOT ladder.

    This is the PennyLane equivalent of Qiskit's
    TwoLocal(rotation_blocks=["ry","rz"], entanglement_blocks="cx").

    It is applied on top of the Hartree-Fock reference state so that the
    comparison with UCCSD isolates the CIRCUIT rather than the starting
    point. Note it does NOT conserve particle number - nothing in its
    structure knows about chemistry.
    """
    hf = qml.qchem.hf_state(electrons, n_qubits)
    n_params = 2 * n_qubits * (reps + 1)

    def ansatz(params, wires):
        qml.BasisState(hf, wires=wires)
        p = params.reshape(reps + 1, n_qubits, 2)
        for layer in range(reps):
            for q in range(n_qubits):
                qml.RY(p[layer, q, 0], wires=q)
                qml.RZ(p[layer, q, 1], wires=q)
            for q in range(n_qubits - 1):
                qml.CNOT(wires=[q, q + 1])
        for q in range(n_qubits):
            qml.RY(p[reps, q, 0], wires=q)
            qml.RZ(p[reps, q, 1], wires=q)

    return ansatz, n_params
```

Why start from the Hartree-Fock state

It makes the comparison fair. If UCCSD started from HF and the hardware-efficient circuit started from $|0000\rangle$, you would be measuring the reference state as much as the circuit. Holding the starting point fixed isolates the variable you actually care about.

Design your experiments so that exactly one thing differs. It is the difference between a result and an anecdote.

Step 6.2 — Four conditions across the curve

Listing 6.2 — the comparison

```
def compare(grid=None):
    """UCCSD vs hardware-efficient, warm and cold, across the whole curve."""
    if grid is None:
        grid = make_grid()

    n = len(grid)
    out = {k: np.zeros(n) for k in
           ["e_fci", "uccsd", "hea_warm", "hea_cold", "hea_deep"]}
```

```

p_uccsd = p_heal = None

for i, r in enumerate(grid):
    _, out["e_fci"][i], _ = classical_energies(r)
    H, nq = qubit_hamiltonian(r)

    anz_u, np_u = build_uccsd(nq)
    if p_uccsd is None:
        p_uccsd = pnp.zeros(np_u, requires_grad=True)
    out["uccsd"][i], p_uccsd, _ = run_vqe(H, nq, anz_u, p_uccsd)

    anz_h, np_h = build_heal(nq, reps=2)
    if p_heal is None:
        rng = np.random.default_rng(7)
        p_heal = pnp.array(rng.normal(0, 0.05, np_h), requires_grad=True)
    out["heal_warm"][i], p_heal, _ = run_vqe(H, nq, anz_h, p_heal,
                                              max_iter=600)

    out["heal_cold"][i] = _best_of(H, nq, anz_h, np_h, seed0=1000)

    anz_d, np_d = build_heal(nq, reps=4)
    out["heal_deep"][i] = _best_of(H, nq, anz_d, np_d, seed0=2000)

out["grid"] = grid
return out

```

Step 6.3 — The diagnostic that separates the two failure modes

Listing 6.3 — the diagnostic, with its control

```

def diagnose(r=2.5, n_restarts=8, depths=(1, 2, 3, 4, 6)):
    """Why does the hardware-efficient ansatz fail at long bond length?

    Two explanations, which are NOT mutually exclusive:
    - OPTIMISER TRAPPING -> many restarts pile up on one non-optimal value
    - EXPRESSIBILITY      -> even the BEST restart cannot reach the answer

    Do not use the spread (peak-to-peak) as the test: a single lucky restart
    that escapes a dominant basin inflates it and inverts the verdict.
    """
    _, e_fci, _ = classical_energies(r)
    H, nq = qubit_hamiltonian(r)

    energies = []
    for s in range(n_restarts):
        anz, npar = build_heal(nq, reps=2)
        rng = np.random.default_rng(100 + s)
        p0 = pnp.array(rng.normal(0, 0.5, npar), requires_grad=True)
        e, _, _ = run_vqe(H, nq, anz, p0, max_iter=800)
        energies.append(e)

    energies = np.array(energies)
    best = energies.min()
    values, counts = np.unique(np.round(energies, 6), return_counts=True)
    modal_value, modal_count = values[counts.argmax()], counts.max()

    print(f"  modal cluster          : {modal_value:.6f} Ha "
          f"({modal_count}/{n_restarts} restarts)")
    print(f"  best over all restarts: {best:.6f} Ha")

    trapped = modal_count >= 0.5 * n_restarts and modal_value > best + 1e-6
    print("  ->", "dominant attractor basin: most restarts are TRAPPED"

```

```

        if trapped else "no dominant basin: restarts explore freely")

# Random restarts bound what random SEARCH finds - not what the circuit
# can EXPRESS. To separate those you must reach the same geometry by a
# path the optimiser can actually follow. Continuation does that.
e_cont = continuation(r, reps=2)
print(f" same circuit via continuation : {e_cont:.6f} Ha")

if abs(e_cont - e_fci) < CHEMICAL_ACCURACY:
    print(" -> NOT an expressibility limit. The solution is inside this")
    print(" circuit's reach; random initialisation never finds it.")

def continuation(r_target=2.5, reps=2, start=0.30, step=0.05):
    """Reach r_target by walking there from an easy geometry.

    This is the control for the restart experiment. If continuation gets a
    result random restarts cannot, the circuit was always expressive enough
    and the problem is purely where the optimiser starts.
    """
    path = np.round(np.arange(start, r_target + 1e-9, step), 4)
    params = None
    energy = None
    for r in path:
        H, nq = qubit_hamiltonian(float(r))
        anz, npar = build_heh(nq, reps=reps)
        if params is None:
            rng = np.random.default_rng(7)
            params = pnp.array(rng.normal(0, 0.05, npar), requires_grad=True)
        energy, params, _ = run_vqe(H, nq, anz, params, max_iter=600)
    return energy

```

4. Results

4.1 The headline

Condition	Parameters	Max error	Geometries above chemical accuracy
UCCSD, warm-started	3	0.0013 mHa	0 / 43
Hardware-efficient, 2 reps, warm-started	24	0.054 mHa	0 / 43
Hardware-efficient, 2 reps, cold-started	24	24.10 mHa	8 / 43
Hardware-efficient, 4 reps, cold-started	40	0.098 mHa	0 / 43

The cold-started shallow circuit fails, and it fails in a physically meaningful place: **every geometry at $R \geq 1.80 \text{ \AA}$** , and none below it. That is precisely the strongly-correlated region where Phase 5 showed Hartree-Fock diverging — where the ground state stops being one determinant and becomes a genuine mixture.

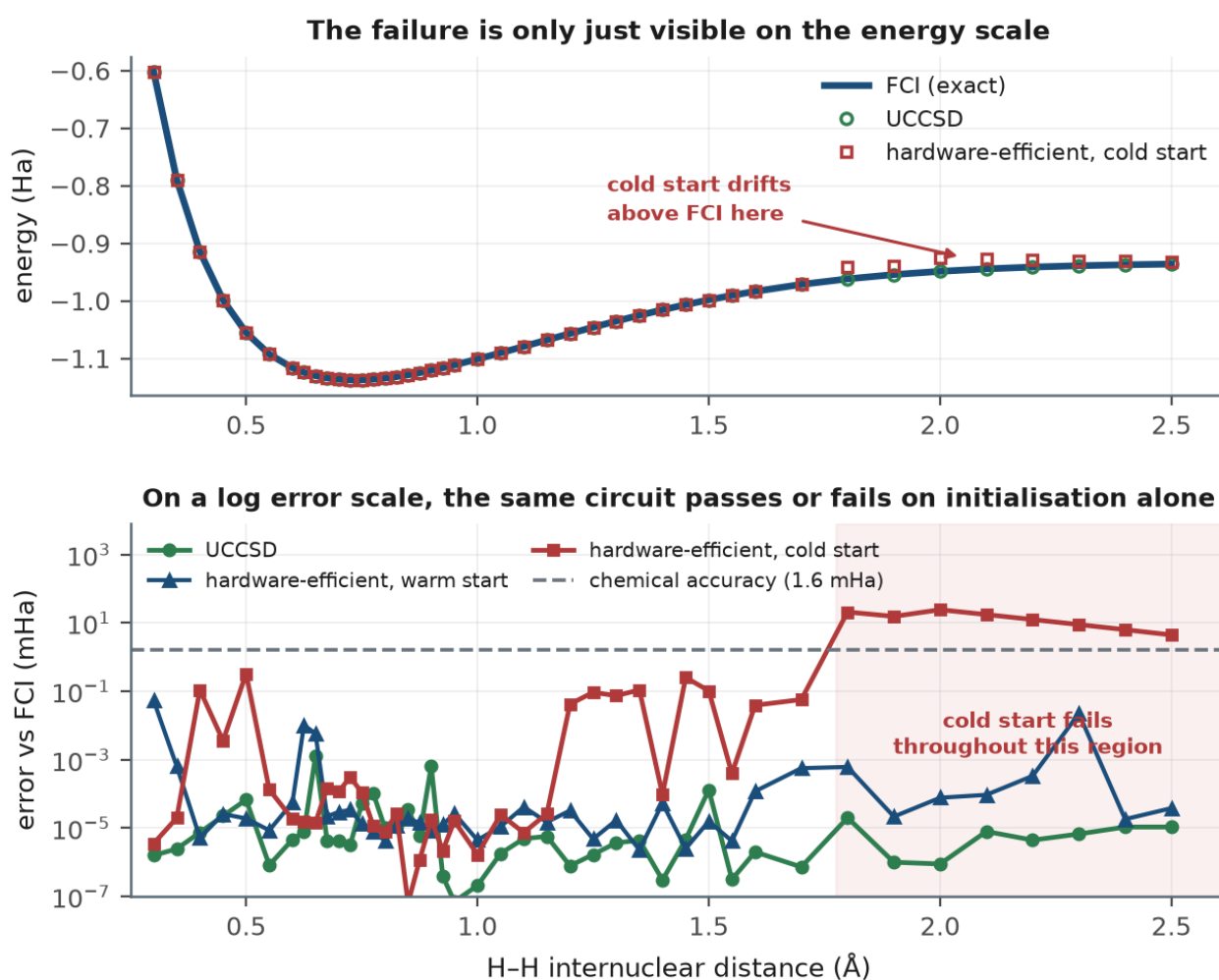


Figure 6.2 — On the energy scale all four are indistinguishable. The log-scale error plot is where the result lives.

4.2 Diagnosing the failure


```

random restarts at r=2.5 Å (FCI = -0.936055 Ha)
restart 0: -0.931639 ( 4.416 mHa)
restart 1: -0.931639 ( 4.416 mHa)
restart 2: -0.931639 ( 4.416 mHa)
restart 3: -0.931639 ( 4.416 mHa)
restart 4: -0.933849 ( 2.206 mHa)
restart 5: -0.931639 ( 4.416 mHa)
restart 6: -0.931639 ( 4.416 mHa)
restart 7: -0.931639 ( 4.416 mHa)

distinct minima found : 2
modal cluster          : -0.931639 Ha (7/8 restarts)
best over all restarts: -0.933849 Ha (2.206 mHa)
-> dominant attractor basin: most restarts are TRAPPED

best over 8 random restarts : -0.933849 Ha (2.206 mHa)
same circuit via continuation : -0.936055 Ha (0.000 mHa)
-> NOT an expressibility limit. The solution is inside this
    circuit's reach; random initialisation simply never
    finds it. The failure is the optimisation landscape.

circuit depth at r=2.5 Å (best of 3 cold starts each)
reps=1 (16 params): -0.931639 ( 4.416 mHa)
reps=2 (24 params): -0.931639 ( 4.416 mHa)
reps=3 (32 params): -0.935728 ( 0.327 mHa) < chemical accuracy
reps=4 (40 params): -0.936009 ( 0.046 mHa) < chemical accuracy
reps=6 (56 params): -0.935811 ( 0.244 mHa) < chemical accuracy

```

Diagnosing why the hardware-efficient ansatz fails at 2.5 Å

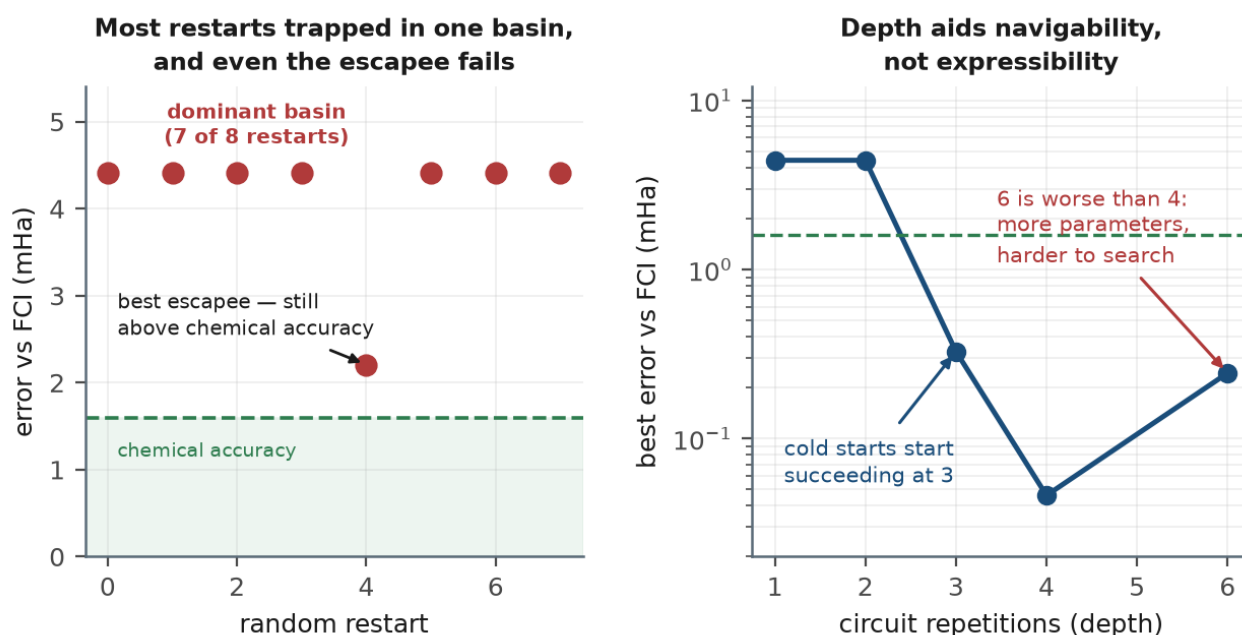


Figure 6.3 — The diagnostic, both panels.

Three separate findings come out of this, and all three matter.

Finding 1 — The obvious statistic gets it backwards

Seven of eight restarts converge to **exactly** -0.931639 Ha. Random local minima do not agree to six decimal places, so that is a single dominant attractor basin catching almost every start. The eighth reaches -0.933849 Ha, which is **lower** — so -0.931639 is not a ceiling, merely a

very sticky basin.

Extending to thirty restarts maps the landscape: thirteen distinct minima, roughly twenty in the -0.9316 basin, the remainder around -0.9338 , and a best of -0.933866 Ha (2.19 mHa). **Not one of thirty random starts reached chemical accuracy.**

Peak-to-peak spread is the wrong test

The spread across the eight restarts is 2.21 mHa, which looks like scatter and reads as "optimiser trapping". But a single lucky escapee dominates a peak-to-peak statistic, and the same number would appear whether or not a ceiling existed.

The two measurements that actually carry information are the **size of the modal cluster** — how sticky the landscape is — and the **best energy over all restarts** — how good the circuit can get when the optimiser cooperates. Listing 6.3 reports both, and was rewritten after this run for exactly that reason.

Even those two are not enough on their own. The decisive experiment is Finding 2.

Finding 2 — It is the landscape, not the circuit

Thirty random restarts of the two-repetition circuit at $2.5 \text{ \AA}$ never beat 2.19 mHa. It would be natural to conclude the circuit cannot express the ground state at that geometry. **That conclusion is wrong**, and one number disproves it:

Same circuit, same geometry ($2.5 \text{ \AA}$), 24 parameters	Best energy	Error vs FCI
30 cold starts, random initialisation	-0.933866 Ha	2.189 mHa
1 warm start, continued along the curve	-0.936055 Ha	0.0000 mHa

Warm-starting reaches the exact FCI energy at a depth where thirty random starts all failed. The solution was inside the two-repetition circuit's reach the whole time — **there is no expressibility limit here at all**. The failure is entirely a property of the optimisation landscape and where you enter it.

Warm-starting works as **continuation**. At $0.30 \text{ \AA}$ the problem is easy; each subsequent geometry is a small perturbation of the last; and the optimiser is carried into the correct basin at $2.5 \text{ \AA}$ along a path that never required it to solve the hard problem from a cold start. This is the same idea as homotopy or numerical continuation in classical optimisation, and it costs nothing to implement.

Seeing it: the surface underneath the result

The dissociation curve is energy against one variable, so it is a curve and drawing it in three dimensions would add nothing. But there **is** a genuine surface here — the energy as a function of both the geometry R and the ansatz parameter θ — and plotting it makes the whole argument visible at once. `landscape.py` computes it in about 25 seconds.

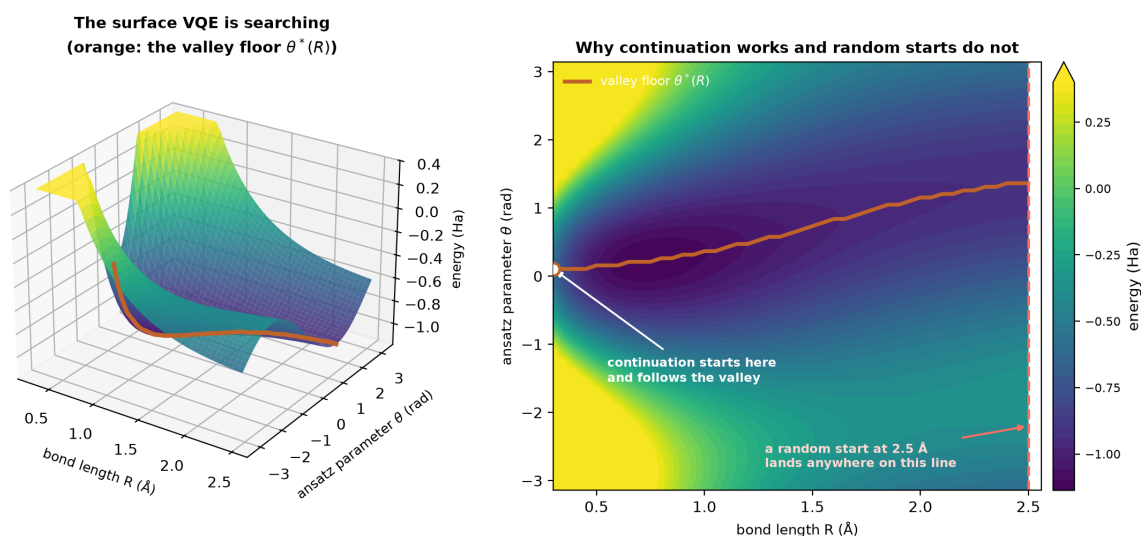


Figure 6.4 — The optimisation landscape $E(R, \theta)$. The orange line is the valley floor: the optimal θ at each geometry.

The valley floor drifts smoothly from $\theta^* = 0.105$ rad at 0.30 Å to 1.361 rad at 2.50 Å — it moves by 1.257 radians across the scan. Continuation enters that valley where the problem is easy and rides it out to full dissociation, which is why it never has to solve the hard problem at all. A random start at 2.5 Å is a blind guess somewhere along the vertical line on the right-hand panel, with no reason whatsoever to land near $\theta^* = 1.36$.

The general lesson

When a parameterised problem is hard at one setting and easy at another, and the two are connected by a continuous path, walk the path. This is a standard technique in classical numerical optimisation — homotopy, or numerical continuation — and it applies to variational quantum algorithms unchanged.

For this project it converts a circuit that fails at eight geometries into one that fails at none, for three lines of code and no extra circuit depth.

Finding 3 — Depth helps by reshaping the landscape, not by adding reachable states

Given Finding 2, the depth study needs re-reading. Best-of-three cold starts gives 4.42 mHa at one and two repetitions, 0.33 mHa at three, 0.046 mHa at four, and 0.24 mHa at six. The natural reading — more depth means more expressibility — is not available, because two repetitions were already expressive enough.

What depth actually buys is **navigability**. At three repetitions, thirty restarts still put twenty-three in the same -0.931639 basin, but one reaches 0.013 mHa. The good basin has not appeared; it has become findable. And the improvement is **not monotonic** — six repetitions are worse than four, because fifty-six parameters make the landscape harder to search faster than the extra freedom helps. That is the barren-plateau trade-off at a scale small enough to watch.

This closes a loop opened in Phase 4

Phase 4 measured warm-starting as worth only 4.9% of optimiser iterations for UCCSD and concluded it looked unimportant. Here the same technique is the difference between a curve that fails chemical accuracy at eight geometries and one that does not.

The lesson is not "warm-starting is good" but something sharper: **techniques that look negligible on an easy problem can be load-bearing on a hard one.** Measure on the problem you actually have.

5. Writing this up

This is a real result and deserves to be reported as one. A defensible summary:

A claim you can defend

"A hardware-efficient ansatz at two repetitions reproduces FCI for H_2 to within chemical accuracy at every geometry up to $1.75 \text{ \AA}$, and fails by up to 24 mHa beyond it when randomly initialised. Thirty random restarts at $2.5 \text{ \AA}$ find thirteen distinct minima and never reach chemical accuracy, the best being 2.19 mHa . The failure is nevertheless not an expressibility limit: the same circuit, warm-started by continuation along the dissociation coordinate, reproduces FCI exactly at that geometry. Increasing depth improves cold-start results non-monotonically — three and four repetitions reach chemical accuracy, six do not improve on four — which is consistent with depth improving the navigability of the landscape rather than enlarging the reachable set. For this system, initialisation strategy dominates circuit depth."

Note what makes that paragraph defensible: every clause is something you measured, the mechanism is distinguished from the symptom by an experiment you designed, and the limitations are stated rather than glossed. That is the difference between a project write-up and a tutorial reproduction.

6. Other directions

If you want to keep going, in rough order of effort against insight:

Extension	Effort	What you would learn
Larger molecule (LiH, BeH_2)	Low	12 spin orbitals means 12 qubits and a much larger excitation space. Active-space selection becomes a real decision.
Bigger basis (6-31G, cc-pVDZ)	Low	Watch the 17% experimental disagreement from Phase 5 shrink. Every function already takes <code>basis</code> as an argument.
Shot noise	Medium	Set finite shots on the device and watch the curve acquire scatter. Quantifies how many measurements chemical accuracy actually costs.
Noise model	Medium	Add a depolarising channel and measure the upward bias. This is where the hardware-efficient ansatz's shallowness finally pays for itself.
Real hardware	High	Queue times, transpilation, error mitigation. The roadmap's warning stands: error mitigation is a project in itself, not a footnote.
ADAPT-VQE	High	Build the ansatz operator by operator, chosen by gradient magnitude. Directly addresses the expressibility-versus-depth trade-off you measured here.

7. What was done in this phase

- **Turned an untested assertion into a measured result.** Phase 3 claimed ansatz choice matters; this phase quantified exactly where, by how much, and why.

- **Designed a controlled experiment** — same Hamiltonian, same optimiser, same reference state, one variable changed — which is what separates a result from an anecdote.
- **Found the failure boundary at $R \geq 1.80 \text{ \AA}$** , coinciding with the onset of static correlation identified in Phase 5. The two findings explain each other.
- **Learned that the obvious diagnostic is wrong.** Peak-to-peak spread across restarts is dominated by a single lucky escapee and reports the same value whether or not a ceiling exists. Modal cluster size and best-over-restarts carry the information instead.
- **Learned that even those are not sufficient.** Thirty random restarts failing to reach chemical accuracy looks like proof of an expressibility limit and is not — it bounds what random *search* finds, not what the circuit can *express*. The control experiment is continuation, and it inverted the conclusion.
- **Established that the failure is entirely the optimisation landscape:** the same 24-parameter circuit that thirty random starts cannot get within 2.19 mHa of FCI reproduces it exactly when reached by continuation from equilibrium.
- **Re-read the depth study in that light:** depth improves navigability rather than expressibility, and does so non-monotonically — six repetitions are worse than four.
- **Revised Phase 4's conclusion** that warm-starting was of marginal value. On this problem it is the dominant factor.
- **Produced a claim worth defending**, with mechanism, evidence and limitations stated.

8. Final checklist

1. `compare_ansatz.py` runs and saves `data/ansatz_comparison.npz`.
2. You reproduce the failure of the cold-started 2-repetition circuit at long bond length.
3. The restart diagnostic shows a dominant modal cluster, and you can say what that does and does not imply.
4. You have run the continuation control and seen it beat every random restart at the same depth.
5. You can state why thirty failed restarts do **not** establish an expressibility limit.
6. Your write-up states the mechanism, not just the observation.
7. You can explain why warm-starting mattered here and barely mattered in Phase 4.

You are done

Seven phases: an environment, a classical oracle, a qubit Hamiltonian you control end to end, a verified VQE, a potential energy surface, three physical constants, and an original experiment with a defensible result.

The thing worth remembering is the shape of it — build the reference before the thing it checks, verify at every stage, and measure rather than assume. That structure transfers to problems far larger than H_2 .